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The relevance of placenta MRI diffusion-derived vessels density and morphological features in diagnosing placenta accreta spectrum

Tao Lu^{1*†}, Dian-Qi Yao², Ben-Heng Xiao², Wei Sheng³, Meining Chen³ and Yi Xiáng J. Wáng^{2*†}

Abstract

Objectives MRI morphological features are postulated to represent PAS (placenta accreta spectrum), while diffusion-derived vessel density (DDVD) computed from a simple diffusion MRI protocol is a new imaging biomarker representing perfusion. This study aims to explore the relevance of DDVD and MRI morphological features in diagnosing PAS.

Methods We enrolled 86 patients with PAS and 40 pregnant women without PAS. Each case underwent intravoxel incoherent motion (IVIM) MRI sequence with 11 b -values. DDVD was the signal difference between $b=0$ and $b=50$ s/mm² images. DDVD_r of the placenta or the placental-myometrial interface was calculated as: [lower 1/3 placenta DDVD]/ [upper 2/3 placenta DDVD], respectively. Differences of DDVD_r in non-PAS pregnancies and in mild or severe PAS patients were explored using Kruskal-Wallis H-test. χ^2 test was used to assess the predictive ability of MRI morphological features in diagnosing all PAS and severe PAS. Receiver operating characteristics (ROC) analysis was used to estimate the discriminative ability of DDVD_r and MRI morphological features for all PAS and severe PAS.

Results DDVD_r(placenta) and DDVD_r(interface) were significantly higher in patients with severe PAS (median, 1.194 and 1.230) than those in non-PAS (1.024 and 1.016, both $p < 0.001$). T2 dark bands and abnormal intraplacental vascularity were identified as predictors for all-inclusive PAS and severe PAS. DDVD_r(interface) was significantly higher in patients manifested with T2 dark bands and abnormal intraplacental vascularity (median, 1.242 vs. 1.116, 1.230 vs. 1.126 $p < 0.05$). When DDVD_r(interface) and the MRI morphological features were combined for predicting all-inclusive PAS and severe PAS, the AUC reached 0.894 and 0.916, respectively.

Conclusion The combination of placenta MRI diffusion-derived vessels density and qualitative can be used for the accurate prediction of PAS.

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Keywords Placenta, Placenta accreta spectrum disorders, Diffusion-weighted MRI, Diffusion-derived vessel density, Perfusion

Placenta accreta spectrum (PAS) is a pathologic condition when the placental villi abnormally implanted into the myometrium due to the defect of the endometrial-myometrial interface, leading to failure of normal decidualization and normal placental detachment at the time of delivery [1, 2]. The failure of placental separation from uterus subsequently puts women at higher risk of postpartum hemorrhage, making PAS a major cause of maternal morbidity and mortality [3]. The incidence of PAS has kept increasing during the past 50 years. It was 1 in 333 of all deliveries by the end of 2017 in the United States and becomes a health concern for pregnant women especially with a history of previous cesarean delivery and placenta previa in an ongoing pregnancy [4]. The increased use of in vitro fertilization (IVF) also contributes to the increase of PAS. It was reported that IVF rates were twice among women with PAS than those without [3]. The severity of PAS is classified as a mild form (placenta accreta) and severe forms (placenta increta and placenta percreta). A recent study debated the traditional description of placenta percreta and found no histopathologic evidence of transmural villous tissue invasion to the uterine serosa and /or into the pelvis [5]. Instead, placenta percreta was the consequence of a uterine rupture, dehiscence or adhesion mainly between the anterior lower uterine segment and the posterior wall of the bladder [5].

Ultrasonography (US) is the main diagnostic imaging modality for PAS, while MRI is increasingly used in conjunction with US in suspected cases or in surgical planning as MRI more clearly depicts the topography and depth of PAS. While perfusion remains an important marker for tissue viability, IVIM has been proposed as a method that measures blood flow in the fetal capillaries and intervillous spaces in the placenta [6, 7]. However, previous studies showed conflicting results. Using Intravoxel incoherent motion (IVIM), León et al. showed IVMI-f was significantly higher in patients with PAS [8]. Similarly, Bao et al. showed IVIM-f and IVIM-D* were significantly higher in the accreta lesions [9]. However, Lu et al. showed IVIM-D was significantly higher in PAS patients while IVIM-f did not differ between patients with and without PAS [10]. Wang et al. proposed the concept of MRI derived DDVD (diffusion derived vessel density) to reflect in vivo tissue vessels density and suggested its potential in evaluation of perfusion of different tissue and different pathology status [11]. Recently, Lu et al. adopted DDVD to evaluate the placental perfusion in PAS patients and found DDVD was significantly higher in PAS patients [12]. Although MRI morphological features have been suggested to diagnose PAS, the

relationship between MRI morphological features and functional parameters has not been fully explored. PAS happens initially in the placental-myometrial interface; however, the perfusion of the placental-myometrial interface is yet to be determined. The goal of this study was to explore the association between MRI morphological features and DDVD in the placental-myometrial interface, and to compare the diagnostic performances of MRI morphological features and DDVD in diagnosing PAS and severe PAS.

Materials and methods

Patients

This study was approved by our institutional review board and informed consent was obtained from each study participant. Data were collected during November 2018 and May 2023 on a 1.5T scanner (Aera, Siemens Healthineers, Erlangen, Germany). Women with a singleton pregnancy, at least one clinical risk factor or ultrasound signs suggestive of PAS were included. We excluded patients with any other maternal diseases or suspected placental insufficiency. Patients with inadequate surgical or pathologic information, or severe artifacts on MR images were also excluded (Fig. 1). Macroscopic features during surgery and pathological examination of the hysterectomy specimens, or placental tissues obtained from sites of invasion were recorded. FIGO classification was used to grade PAS into grade 1 (abnormally adherent placenta, placenta creta), grade 2 (abnormally invasive placenta, placenta increta) and grade 3 (abnormally invasive placenta, placenta percreta) [13]. Patients with severe PAS were defined as those with FIGO grade 2 or 3 PAS, corresponding with placenta increta or percreta, respectively. Patients with mild PAS were defined as those with FIGO grade 1 PAS, corresponding with placenta accreta.

MR protocol

We acquired placenta MRI data on our cohort with a previously reported IVIM sequence [12]. Patients were placed in a supine position and the following sequences were acquired in axial, coronal, and sagittal planes perpendicular to the placenta-myometrium interface including half-Fourier acquisition single-shot turbo spin echo (HASTE), true fast imaging in steady-state precession (TRUFISP) and 3D-volumetric interpolated breath-hold examination (3D-VIBE). For diffusion weighted imaging, single-shot spin-echo type echo-planar sequence imaging was acquired in the axial planes and the following parameters were applied: TR = 5200ms, TE = 83ms, NEX(number of excitation) = 2, FOV = 390 mm, slice

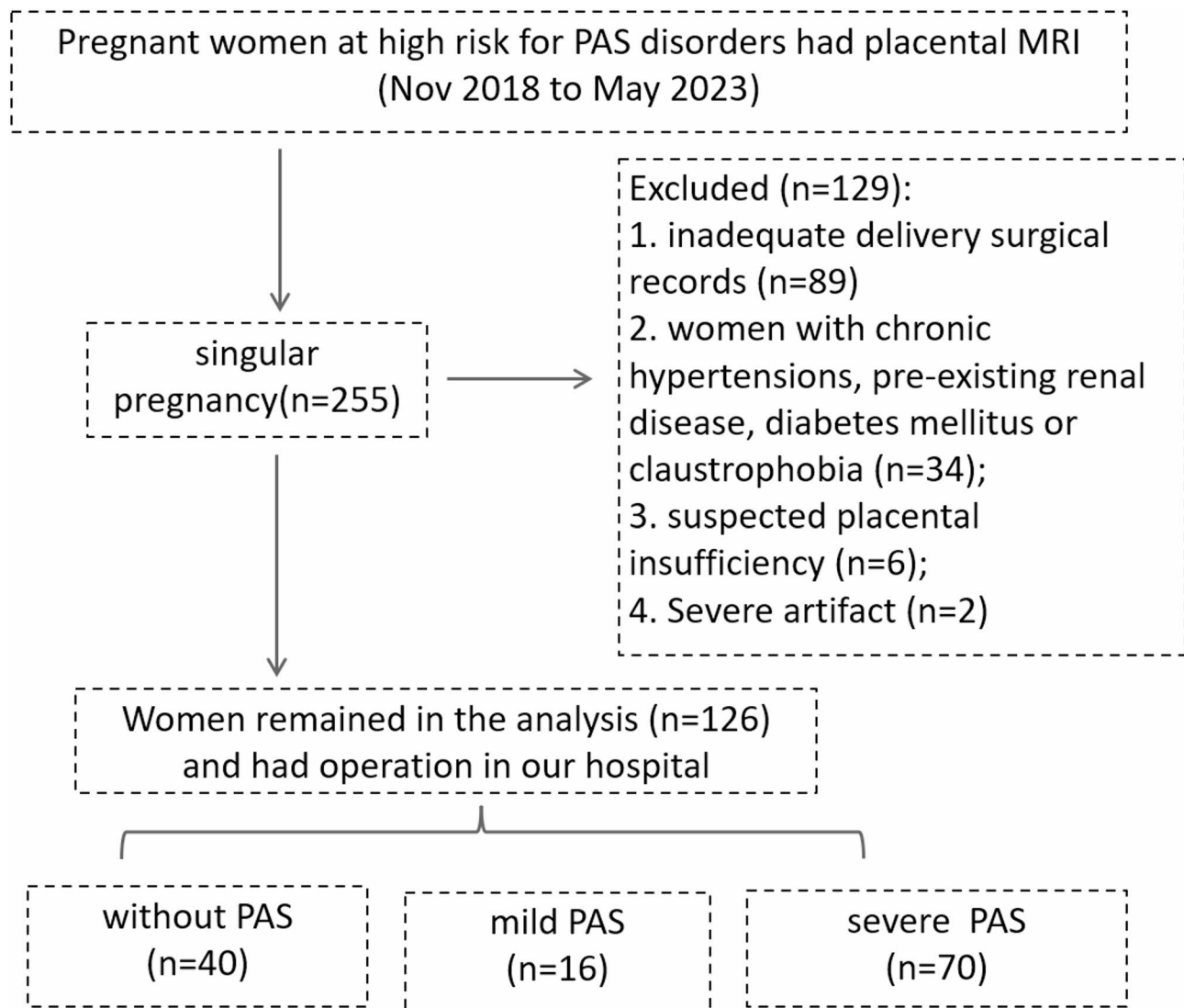


Fig. 1 Study design flowchart

thickness = 5 mm, matrix = 192×120 , parallel imaging acceleration factor = 2, b -values ranging from 0 to 1600 s/mm^2 ($b = 0, 50, 100, 150, 200, 400, 600, 800, 1000, 1200$, and 1600 s/mm^2), scan duration 7 min 29 s. Three-motion probing gradient directions were applied for each b -value. According to the concept of DDVD, its calculation requires data with $b = 0$ image and a low b -value image [11]. Diffusion images with b values of 0 and 50 s/mm^2 were utilized in this study.

DDVD quantification

DDVD measurement was performed with Yingran Image Analytics software Diffeye version 1.0 (Yingran Medicals, Hong Kong SAR, China), and DDVD was calculated using the following equations:

$$\text{DDVD} = \frac{\text{Sb0}}{\text{ROI}_{\text{area0}}} - \frac{\text{Sb50}}{\text{ROI}_{\text{area50}}} [\text{unit: arbitrary unit (au)/pixel}] \quad (1)$$

where denote upper 2/3 placenta refer to the number of pixels in the selected region-of-interest (ROI) on $b = 0 \text{ s/mm}^2$ and $b = 50 \text{ s/mm}^2$ images, respectively. Sb0 refers to the measured signal intensity within the ROI of when $b = 0 \text{ s/mm}^2$, and Sb50 refers to the measured signal intensity within the ROI of when $b = 50 \text{ s/mm}^2$, thus $\text{Sb}/\text{ROI}_{\text{area}}$ equate to the mean signal intensity within the ROI [14]. Two ways to define ROI were tested, the full placenta thickness approach and the placental-myometrial interface approach, as illustrated in Fig. 2. ROIs were drawn independently by two radiologists with five and ten years of experience in obstetric imaging, respectively. To minimize the scaling factors as absolute MR signal intensity is influenced by various factors, including B1 spatial inhomogeneity, coil loading, receiver gain, etc.,

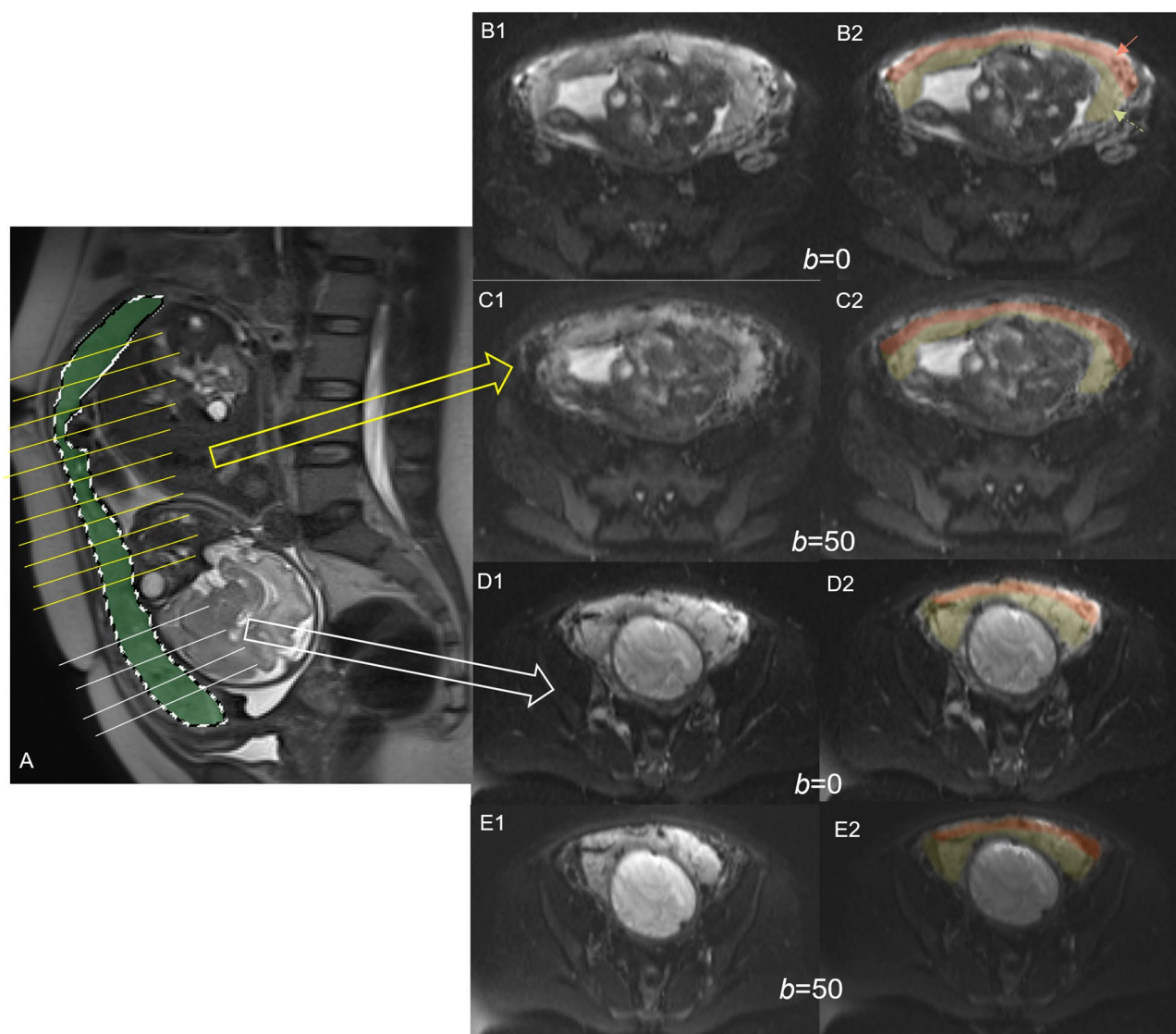


Fig. 2 An example of region-of-interest (ROI) placement for placenta on diffusion-weighted (DW) images. **A** yellow lines denote upper 2/3 placenta sections and white lines denote lower 1/3 placenta sections. **B** and **C** one representative section from upper 2/3 placenta with b -value = 0 and 50 s/mm². **D** and **E** one representative section from lower 1/3 placenta with b -value = 0 and 50 s/mm². ROIs are shown on B2, C2, D2 and E2. Pink ROIs (pink arrow) represent placental-myometrial interface. Total placenta ROI includes the yellow ROI (yellow arrow) and the interior portion of the pink ROIs

the lower 1/3 part and the upper 2/3 part of the placenta and the placental-myometrial interface were delineated, respectively. The ratios of lower 1/3 part to upper 2/3 part of the ROI were taken as:

$$\text{DDVDr (placenta)} = \frac{\text{lower 1/3 part DDVD (placenta)}}{\text{upper 2/3 part DDVD (placenta)}} \quad (2)$$

$$\text{DDVDr (interface)} = \frac{\text{lower 1/3 part DDVD (interface)}}{\text{upper 2/3 part DDVD (interface)}} \quad (3)$$

Qualitative PAS morphological feature analysis

All MR image readings were performed by a radiologist (with 8 years of experience in obstetric imaging) who was

blind to ultrasound diagnosis, surgical and pathological findings. The reviewer read the MR images for presence of the following 12 features: T2 dark bands, placental heterogeneity, abnormal intraplacental vascularity, placental cervical protrusion sign, focal exophytic mass, placental bulge, placental recess, abnormal vascularization of the placental bed, parametrial vessel sign, bladder tenting, bladder vessel sign and myometrial interruption [15–17] (Fig. 3).

Statistical analysis

Continuous variables with a normal distribution are presented with means $\pm$ standard deviation (SD), with a non-normal distribution were presented with median (range),

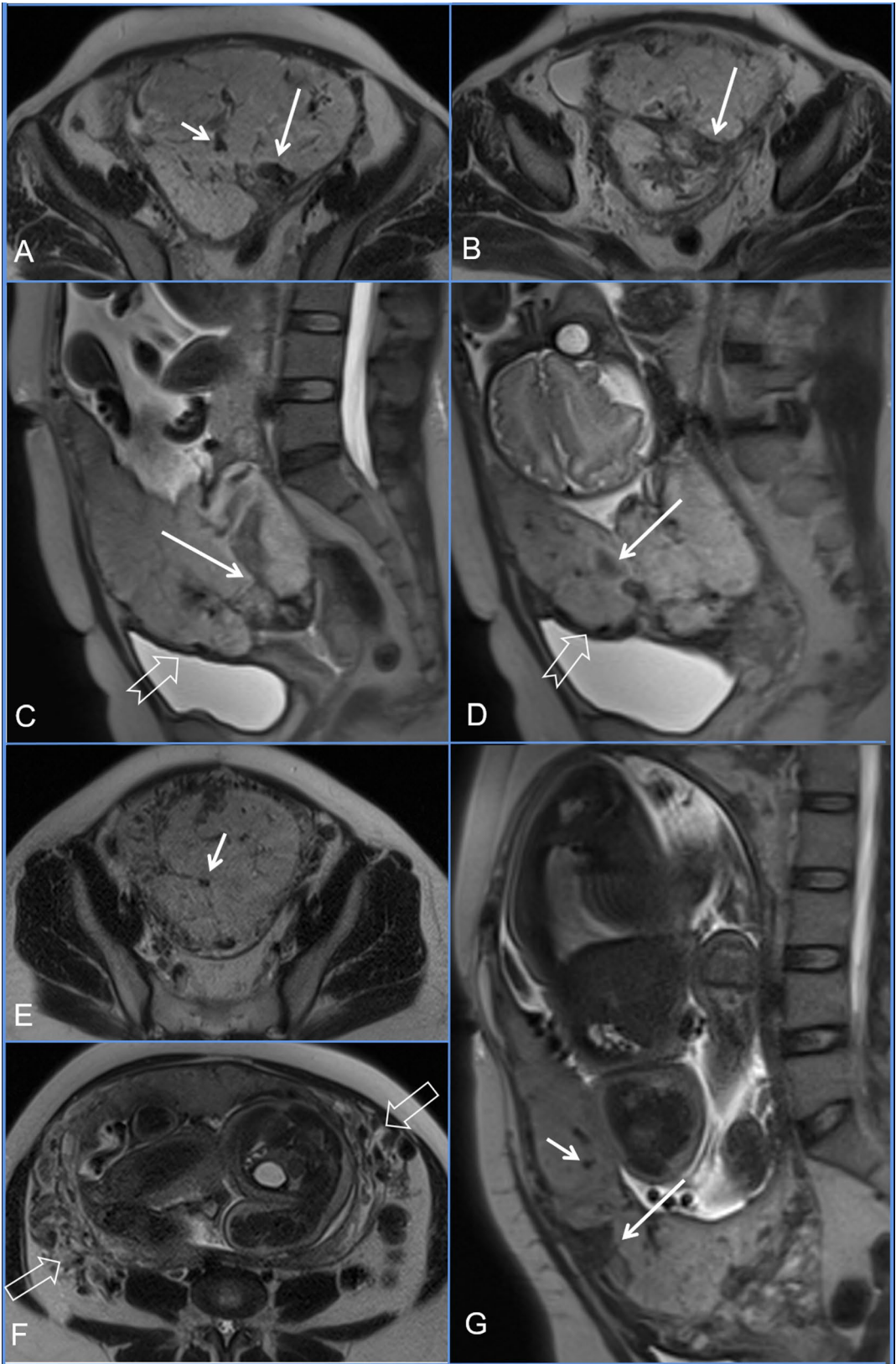


Fig. 3 (See legend on next page.)

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Fig. 3 Presentative image of PAS morphological signs. **A–D** A 35-year-old woman who had one prior cesarean section presented with complete placenta previa and concerns regarding PAS. The patient was at 33 weeks of gestation with placenta percreta. (**A, B**) axial, and (**C, D**) sagittal half-Fourier acquisition single-shot turbo spin echo images. T2 dark bands (long white arrow), abnormal intraplacental vascularity (short white arrow), placental heterogeneity, placental cervical protrusion sign, focal exophytic mass (thick hollow arrow), placental bulge (thick hollow arrow), placental recess are observed. **E–G** A 33-year-old woman who had one prior cesarean section presented with complete placenta previa and concerns regarding PAS. The patient was at 36 weeks of gestation with placenta percreta. (**E, F**) axial and (**G**) sagittal half-Fourier acquisition single-shot turbo spin echo images. T2 dark bands (long white arrow), abnormal intraplacental vascularity (short white arrow), abnormal vascularization of the placental bed and parametrial vessel sign (thick white arrow) are observed

respectively, and categorical variables were presented with absolute numbers (proportion, %). The Mann-Whitney U-test and χ^2 test were used to compare the clinical features between groups. χ^2 test was used to assess the predictive ability of MRI morphological features in diagnosing all-inclusive PAS and severe PAS. Logistic regression analysis was used to identify the independent risk factors predictive of all-inclusive PAS and severe PAS. The Kruskal-Wallis H-test was used to compare the difference of DDVDr of the placenta and the placental-myometrial interface in nonPAS and mild or severe PAS. Receiver operating characteristics (ROC) analysis was used to estimate the discriminative ability of DDVDr and MRI morphological features for all-inclusive PAS and severe PAS. P values < 0.05 were considered statistically significant and analyses were conducted using SPSS 21.0 (IBM Inc).

Results

The maternal characteristics of the 126 patients included in this study are shown in Table 1. The mean maternal age was 32 years, ranging from 22 to 45 years, and the mean gestational age at examination was 31 weeks, ranging from 22 to 38 weeks for all the included patients. Five patients had hysterectomy, and the other 121 patients had their uterus preserved. According to FIGO classification, a total of 68.25% (86/126) of patients were classified as having PAS, including 16 with mild PAS (FIGO grade 1 PAS) and 70 with severe PAS (56 with FIGO grade 2 PAS and 14 with FIGO grade 3 PAS). Patients with severe PAS, compared to non-PAS patients, had an older maternal age (33.27 ± 3.98 vs. 29.15 ± 4.11 , $p < 0.001$), a higher number of patients older than 35 years (45% vs. 10%, $p = 0.001$), a lower gestational age at delivery (36 vs. 38, $p < 0.001$), a higher frequency of prior cesarean Sect. (67.5% vs. 47.5%, $p = 0.007$), a higher number of prior cesarean sections ($p = 0.014$) and a higher number of prior miscarriage ($p = 0.039$). Patients with severe PAS had the highest frequency of placenta previa and non-PAS had the lowest frequency of placenta previa ($p < 0.001$, Table 1). Patients with mild PAS and non-PAS showed no significant difference in terms of maternal age, gestational age at delivery, frequency of prior cesarean sections, number of prior cesarean sections and prior miscarriage (all $p > 0.05$). The gestational age at MRI

and frequency of prior miscarriage did not differ among the 3 groups (all $p > 0.05$).

DDVDr(placenta) and DDVDr(interface) were significantly higher in patients with severe PAS (median, 1.194 and 1.230) than those in non-PAS (1.024 and 1.016, both $p < 0.001$), respectively. A trend of greater differences was noted for DDVDr (interface) comparison than for DDVDr (placenta) comparison, supporting the usefulness of DDVDr (interface) approach (Fig. 4).

T2 dark bands, placental heterogeneity, abnormal intraplacental vascularity, placental cervical protrusion sign, focal exophytic mass, placental bulge, abnormal vascularization of the placental bed and parametrial vessel sign were associated with all PAS, and the diagnostic performance of the 8 MRI morphological features for PAS were presented in Table 2. T2 dark bands demonstrated an AUC of 0.689 and abnormal intraplacental vascularity demonstrated an AUC of 0.745. In multivariate analysis, T2 dark bands (OR:7.141, 95% CI: 1.745–29.216) and abnormal intraplacental vascularity (OR:6.195, 95% CI: 2.059–18.638) were predictors for all PAS. When compared with patients manifested without T2 dark bands, DDVDr(placenta) and DDVDr(interface) were both significantly higher in patients manifested with T2 dark bands [(1.180 vs. 1.061, $p = 0.006$) and (1.242 vs. 1.116, $p < 0.001$)]. When compared with patients manifested without abnormal intraplacental vascularity, DDVDr(placenta) and DDVDr (interface) were both significantly higher in patients manifested with abnormal intraplacental vascularity [(1.193 vs. 1.045, $p < 0.001$) and (1.230 vs. 1.126, $p < 0.001$)] (Fig. 4). For predicting all PAS, DDVDr (placenta) demonstrated an AUC of 0.761 (95%CI 0.673–0.848) with a cutoff value of 1.06, DDVDr (interface) demonstrated an AUC of 0.797 (95%CI 0.711–0.884) with a cutoff value of 1.10. When DDVDr(placenta), T2 dark bands and abnormal intraplacental vascularity were combined to predict all PAS, an AUC of 0.879 (95% CI 0.810–0.948) was achieved. When DDVDr(interface), T2 dark bands and abnormal intraplacental vascularity were combined to predict all PAS, an AUC of 0.894 (95% CI 0.833–0.955) was achieved (Table 3).

T2 dark bands, placental heterogeneity, abnormal intraplacental vascularity, placental cervical protrusion sign, focal exophytic mass, placental bulge, abnormal vascularization of the placental bed and parametrial

Table 1 Maternal characteristics in the study groups

	Cases without PAS (n = 40)	Mild PAS cases (n = 16)	Severe PAS cases (n = 70)	P value
Age(years)	29.15 ± 4.11	29.44 ± 3.83	33.27 ± 3.98	< 0.001
Less than 35	36 (90%)	15 (93.75%)	44 (62.86%)	0.001
35 or older	4 (10%)	1 (6.25%)	26 (37.14%)	
Gestational age At MRI	31 (30,34)	32 (29.25,35)	31(29,33)	0.462
Gestational age At of delivery	38 (37,39)	36 (36,38)	36 (34,37)	< 0.001
Previous caesarean Section				0.007
yes	19 (47.5%)	10 (62.5%)	54 (77.14%)	
no	21 (52.5%)	6 (27.5%)	16 (22.86%)	
No. of previous caesarean Section				0.014
0	21 (52.5%)	7 (43.75%)	19 (27.14%)	
1	17 (42.5%)	8 (50%)	40 (57.14%)	
2 or more	2 (5%)	1 (6.25%)	11 (15.71%)	
Previous miscarriage				0.092
Yes	26 (65%)	10 (62.5%)	57 (81.43%)	
No	14 (35%)	6 (27.5%)	13 (18.57%)	
No. of Previous miscarriage				0.039
0	14 (35%)	5 (31.25%)	10 (14.29%)	
1	12 (30%)	6 (37.5%)	21 (30%)	
2 or more	14 (35%)	5 (31.25%)	39 (55.71)	
Placenta previa				< 0.001
Yes	15 (37.5%)	12 (75%)	68 (97.14%)	
No	25 (62.5%)	4 (25%)	2 (2.86%)	

Gestational age: in weeks

vessel sign were associated with severe PAS (Table 2). In multivariate analysis, T2 dark bands (OR:6.85, 95% CI, 1.615–29.060) and abnormal intraplacental vascularity (OR:11.93, 95% CI, 3.911–36.391) were predictors for severe PAS. For severe PAS prediction, T2 dark bands demonstrated an AUC of 0.709, abnormal intraplacental vascularity demonstrated an AUC of 0.777 (Table 2). For predicting severe PAS, DDVDr(placenta) demonstrated an AUC of 0.79 (95%CI 0.706–0.874) with a cut-off value of 1.06 and DDVDr(interface) demonstrated an AUC of 0.819 (95%CI 0.737–0.901) with a cutoff value of 1.10. When DDVDr(placenta), T2 dark bands and abnormal intraplacental vascularity were combined to predict severe PAS, an AUC of 0.896 (95% CI 0.830–0.962) was achieved. When DDVDr(interface), T2 dark bands and abnormal intraplacental vascularity were combined to predict severe PAS, an AUC of 0.916 (95% CI 0.860–0.972) was achieved.

Discussion

This study shows DDVDr was significantly higher in severe PAS, T2 dark bands and abnormal intraplacental vascularity were independently associated with all PAS and severe PAS, patients manifested with T2 dark bands or abnormal intraplacental vascularity had significantly

higher DDVDr than patients did not manifest with the 2 features. For predicting all PAS and severe PAS, DDVDr demonstrated higher AUC than MRI morphological features. When DDVDr (interface) and MRI morphological features were combined, the AUC reached 0.894 and 0.916, respectively, in prediction of all PAS and severe PAS.

DDVD can be interpreted as a physiological vessel density per unit tissue area [11], with the signal difference between images when the diffusion gradient is 'off' and images when the diffusion gradient is 'on' [11, 18]. If multiple slices were integrated, it can be concept as the volume of micro-vessels per tissue within the volume. DDVD measure based on this simple principle appears to be useful as a straightforward imaging biomarker. For the liver and spleen DDVD measurement, when the TE is short or modest and TR is long, $DDVD_{liver}$ is approximately similar to $DDVD_{spleen}$ which agrees with the known physiology that liver and spleen have similar amount of blood perfusion [19]. Moreover, when $b = 0$, 2 mm^2/s were used for the DDVD calculation of 26 cases of HCC (hepatocellular carcinoma), we measured a mean $DDVD_{HCC}/DDVD_{liver}$ ratio of 1.42. This value agrees well with the perfusion CT blood volume literature results median ratio of 1.38 [20]. One study of parotid tumor

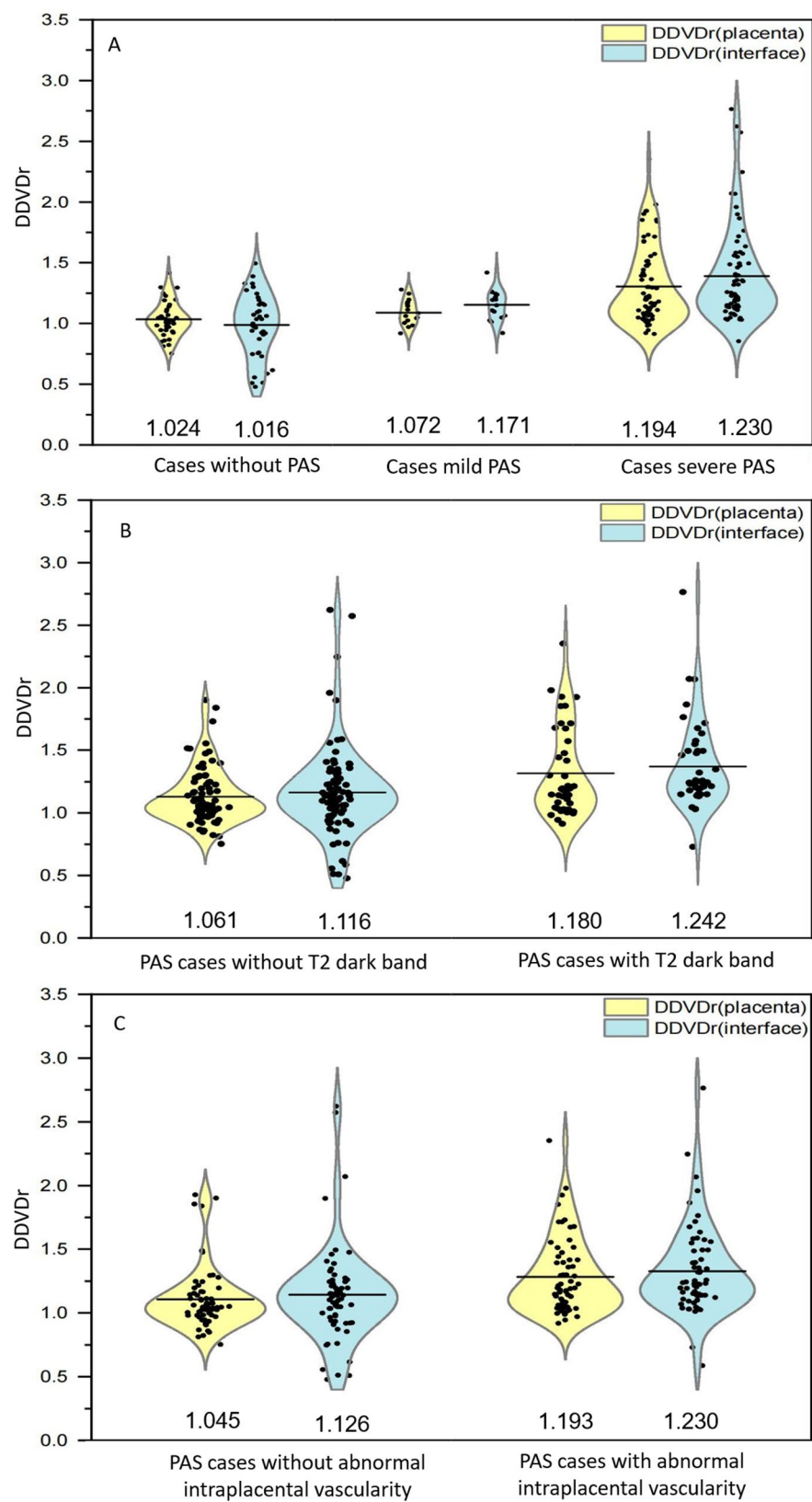


Fig. 4 (See legend on next page.)

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Fig. 4 Box and whisker plots of DDVDr. **A** Sensitivity and diagnostic performance of individual with severe PAS than that in cases without PAS ($p < 0.001$). A trend is noted with mild PAS cases have slightly higher DDVDr than cases without PAS. A trend is noted with DDVDr(interface) showing a greater difference in cases with and without PAS, thus favoring the approach of DDVDr(interface). **B** DDVDr is s higher in cases manifested with T2 dark bands ($p < 0.001$). **C** DDVDr is higher in cases manifested with abnormal intraplacental vascularity ($p < 0.05$). In both B and C, the differences are better shown with the DDVDr(interface) approach. The values noted in the graphs are median values

DDVD also showed parotid tumor DDVD ratio measures were consistent with CT measured blood volume ratios [21]. Lu et al. [22] reported earlier clinical grades rectal carcinoma had a higher DDVDr (tumor to tumor-free rectal wall) than those of the advanced clinical grades (2.245 for grade 0&I, 1.460 for grade II, 1.430 for grade III, 1.130 for grade IV). This is consistent with the biological behaviours of rectal carcinoma. DDVD is a useful parameter for the distinguishing of livers with and without fibrosis, and livers with severer fibrosis tend to have even lower DDVD measurements than those with milder liver fibrosis [11, 23, 24]. Moreover, He et al. [25]. and Li et al. [26]. reported that placenta DDVD as a perfusion biomarker of the placenta allows excellent separation of normal and early preeclampsia pregnancies. Wang et al. [27]. described that endometrial carcinoma with Ki-67 high-proliferation or aggressive histological type had higher DDVD value than those with Ki-67 low-proliferation or non-aggressive histological type. ROC curve analysis shows AUC of 0.873 for distinguishing between Ki-67 low- and high-expression, and AUC of 0.771 for distinguishing between non- aggressive and aggressive histological types. Ni et al. [28] applied DDVD Habitat analysis to separate isocitrate dehydrogenase (IDH) -mutant and IDH-wildtype diffuse gliomas, achieved a AUROC of 0.823. Lu et al. [12] adopted DDVD to evaluate PAS, confirmed the negative association between placenta DDVD and gestational age, and showed DDVDr was significantly higher in patients with PAS [12]. However, the exact correlation between the MRI morphological features and the physiological meaning of DDVD in the setting of PAS has not been explored.

T2 dark bands are irregular, thick bands of low T2 signal intensity originating from the maternal surface of the placenta and crossing the placenta perpendicularly [29–31]. T2 dark bands usually have a size of 6 to 20 mm and represent fibrin deposition from hemorrhage and infarct [32–34]. Abnormal intraplacental vascularity is an uncertain MRI feature for PAS from SAR and ESUR consensus [15]. It refers to prominent, disorganized intraplacental vessels originating from the umbilical cord, running to the deep placental parenchyma, and reaching to maternal surface [35]. Bourgioti et al. concluded that presence of these abnormal vascularity with a diameter ≥ 2 mm were suggestive of PAS [36]. Our study showed patients manifested with T2 dark bands and abnormal intraplacental vascularity had higher DDVDr, suggesting higher blood volume in both the placenta and

the placental-myometrial interface in PAS patients. In PAS patients, the decidual defect promotes the abnormal invasion of the trophoblast and subsequent neovascularization or excessive dilation of the myometrial vasculature [37]. These can explain the T2 dark bands and abnormal intraplacental vascularity demonstrated on prenatal MRI, as well as increasing microvascular perfusion manifested as higher DDVD values. Therefore, the morphological changes of PAS on MRI are correlated with the perfusion disturbance of the placental-myometrial interface.

The accurate grade and topography of PAS are critical to surgical planning and selection of adjuvant therapies including embolization or ligation of the uterine artery, aortic balloon placement and transfusion preparations. Multiple studies demonstrated that maternal outcome can be notably improved if the diagnosis of PAS can be made antenatally as these high risk women can be timely cared by multidisciplinary teams with excellence [38, 39]. For predicting all PAS or severe PAS, MRI morphological features had lower sensitivities but higher specificities. In this study, DDVDr was incorporated to predict all PAS and severe PAS, exhibiting an AUC of 0.894 with 85% sensitivity, 85% specificity and an AUC of 0.916 with 90% sensitivity, 85% specificity, respectively. We noticed that DDVDr(interface) had better performance than DDVDr(placenta), which may indicate the increase of anomalies of the uteroplacental circulation in cases of PAS. MRI morphological features are morphological manifestations of PAS, while DDVD is a measure of micro-vessel perfusion of the placenta. The results from our study suggest the combination of morphology and perfusion of the placenta can be used in the accurate prediction of PAS, especially severe PAS, with an acceptable specificity and improved sensitivity. The analysis of DDVD requires only two b-values (with one being $b = 0 \text{ s/mm}^2$), thus allowing a significantly shorter scanning time than DCE CT/MRI or intravoxel incoherent motion (IVIM) imaging. Compared with DCE imaging, the DDVD protocol does not involve contrast injection, data acquisition is much faster, and data post-processing is also relatively straightforward. Moreover, DDVD is likely to be less susceptible to the T2 shine-through effect compared with ADC and IVIM metrics [19, 40].

Our study for the first time explored the microvascular perfusion in the placental -myometrial interface, but still had some limitations. Firstly, DDVD of this study is calculated from a simplified IVIM protocol without very low

Table 2 Sensitivity and diagnostic performance of individual PAS morphological signs

MRI signs	pts PAS-	pts PAS +	Sensitivity %	Specificity %	PPV %	NPV %	AUC	p
T2 dark bands [¶]	7.5% (3/40)	45.35% (39/86)	45.3	92.5	85.80	62.84	0.689	<0.001
T2 dark bands [#]	7.5% (3/40)	49.30% (35/71)	49.3	92.5	86.80	64.59	0.709	0.001
Placental heterogeneity [¶]	2.5% (1/40)	29.55% (22/86)	25.6	97.5	91.1	56.72	0.615	0.001
Placental heterogeneity [#]	2.5% (1/40)	28.17% (20/71)	28.2	97.5	91.86	57.59	0.628	0.001
Abnormal intraplacental vascularity [¶]	15% (6/40)	63.95% (55/86)	64	85	81.01	70.25	0.745	<0.001
Abnormal intraplacental vascularity [#]	15% (6/40)	70.42% (50/71)	70.4	85	82.44	74.17	0.777	<0.001
Placental cervical protrusion sign [¶]	0% (0/40)	13.95% (12/86)	14	100	100	54.64	0.57	0.009
Placental cervical protrusion sign [#]	0% (0/40)	26.76% (12/71)	16.9	100	100	54.61	0.585	0.004
Focal exophytic mass [¶]	0% (0/40)	27.91% (24/86)	27.9	100	100	58.11	0.64	<0.001
Focal exophytic mass [#]	0% (0/40)	20.99% (22/71)	31	100	100	59.17	0.655	0.007
Placental recess [¶]	0% (0/40)	6.98% (6/86)	7	100	100	51.81	0.535	0.095
Placental recess [#]	0% (0/40)	7.04% (5/71)	7	100	100	51.81	0.539	0.157
Placental bulge [¶]	2.50% (1/40)	20.93% (18/86)	20.9	97.5	89.32	55.21	0.592	0.004
Placental bulge [#]	2.50% (1/40)	25.35% (18/71)	25.4	97.5	91.04	56.65	0.614	0.001
Abnormal vascularization of the placental bed [¶]	10.0% (4/40)	24.42% (21/86)	24.4	95	82.91	55.69	0.597	0.006
Abnormal vascularization of the placental bed [#]	10.0% (4/40)	26.35% (18/71)	25.4	90.5	83.55	56.01	0.602	0.009
Myometrial interruption [¶]	7.50% (3/40)	36.05% (31/86)	36	92.5	82.76	59.11	0.643	<0.001
Myometrial interruption [#]	7.50% (3/40)	33.80% (24/71)	33.8	92.5	81.84	58.29	0.632	0.002
Bladder tenting [¶]	0% (0/40)	2.32% (2/86)	2.3	100	100	50.58	0.512	0.464
Bladder tenting [#]	0% (0/40)	2.82% (2/71)	2.8	100	100	50.71	0.514	0.535
Bladder vessel sign [¶]	0% (0/40)	1.16% (1/86)	1.2	100	100	50.30	0.506	0.685
Bladder vessel sign [#]	0% (0/40)	1.41% (1/71)	1.4	100	100	50.35	0.507	1
Parametrial vessel sign [¶]	0% (0/40)	5.81% (5/86)	5.8	100	100	51.49	0.529	0.143
Parametrial vessel sign [#]	0% (0/40)	7.04% (5/71)	7	100	100	51.81	0.535	0.157

[¶]: Comparison between cases without PAS and cases with PAS. [#]: comparison between cases without PAS and cases with severe PAS

Table 3 Predictive performance of MRI readouts for patients with PAS or severe PAS

Risk factors	AUC	Accuracy (%)	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)
DDVDr (placenta) [¶]	0.761	71	72	70	70.59	71.43
DDVDr (placenta) [#]	0.79	73	76	70	71.70	74.47
DDVDr(interface) [¶]	0.797	74	81	67	71.05	77.91
DDVDr(interface) [#]	0.819	76	85	67	72.03	72.83
Combined model 1 [¶]	0.879	83	84	82	82.35	83.67
Combined model 2 [#]	0.896	85.5	89	82	83.18	88.17
Combined model 2 [¶]	0.894	85	85	85	85	85
Combined model 2 [#]	0.916	87.5	90	85	85.71	89.47

[¶]: all PAS, [#]: severe PAS

Combined model 1: T2 dark bands, Intraplacental vascularity and DDVDr (placenta)

Combined model 2: T2 dark bands, Intraplacental vascularity and DDVDr (interface)

b-values. Theoretically, compared with DDVD computed from *b*=0 and *b*=1 images, DDVD computed from *b*=0 and *b*=50 images as in this study would be contaminated by tissue diffusion and be less reflecting ‘pure micro-vessel density’, thus differs from the initial concept of it [11]. We anticipate that DDVD measures with optimized placenta DWI data acquisition protocol can further improve the sensitivity and specificity for PAS detection. Therefore, DDVD calculated from very low *b*-values should be investigated in future studies. Secondly, we did not perform interobserver differences for MRI morphological features between two readers, but only one experienced radiologist in obstetric imaging conducted MRI feature analysis. In our previous study, substantial interobserver

agreement (*k*>0.7) was observed for T2 dark bands, abnormal intraplacental vascularity, and bladder wall interruption (including ‘tenting’ of the bladder wall). Moderate interobserver agreement (*k*>0.4) was observed for placental heterogeneity, myometrial interruption, placental bulge, and focal exophytic mass [41]. These results supported the reproducibility of the MRI features reviewed by the radiologist in this study. Third, most patients (121, 96.03%) with PAS had their uterus preserved in our study and only intraoperative information, rather than ex vivo gross histopathology, was available in this study. It has been suggested that intraoperative criteria for PAS is acceptable for the confirmative diagnosis [42]. Another potential sampling bias is that all the study

participants had at least one clinical risk factor or ultrasound signs suggestive of PAS, even for control group subjects. This could have potentially increased the MRI false positive rate or higher DDVD measure even in the control group. There is the possibility that, if data were to separate PAS patients and pregnancy without PAS risk factors, the diagnostic performance and AUC would be even better. Multi-center studies with large sample sizes will be required to confirm the initial results described in this study. Computer-aided automated measurement of placenta DDVD will also be another line of research.

In conclusion, our study suggests higher vessel density in severe PAS, which may result in morphological change of the placenta manifested on in MR images. A combination of DDVD and MRI morphological features can help in the more comprehensive and accurate diagnosis of PAS. As DDVD analysis requires only 2 *b*-values, resulting in a rather short scanning time which is acceptable to pregnant women. DDVD analysis can be practically applicable for PAS early detection.

Abbreviations

DDVD	Diffusion-derived vessels density
PAS	Placenta accreta spectrum
IVIM	Intravoxel incoherent motion
ROC	Receiver operating characteristics
AUC	Area under the curve
IVF	In vitro fertilization
US	Ultrasonography
FIGO	The International Federation of Gynecology and Obstetrics
HASTE	Half-Fourier acquisition single-shot turbo spin echo
TRUFISP	True fast imaging in steady-state precession
3D-VIBE	3D-volumetric interpolated breath-hold examination
TR	Repetition time
TE	Echo time
NEX	Number of excitation
FOV	Field of view
SD	Standard deviation
SAR	Society of Abdominal Radiology
ESUR	European Society of Urogenital Radiology
DWI	Diffusion weighted imaging

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Author contributions

T.L. carried out Conception and design of this manuscript, T.L. carried out administrative support, D.Y. and Y.W. carried out provision of study materials, X.H. and W.S. carried out collection and assembly of data, M.C. and W.S. carried out Data analysis and interpretation, and manuscript writing and the final approval of manuscript was carried out by all authors. All authors have read and approved the final version of this manuscript.

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Data availability

The datasets generated during and analyzed during the current study are not publicly available due to PACS system regulated by Sichuan Provincial People's Hospital, but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The ethical review aboard of Sichuan Provincial People's Hospital approved the study. Informed consent was obtained from each of the patient. All methods were performed in accordance with the relevant guidelines and regulations.

Consent for publication

Written consent for the publication of the clinical details of participants was obtained from each of the patient.

Competing interests

The authors declare no competing interests.

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